

SEMINAR III • WEEK 3 • FOUNDATIONS

Numbers & Operations

This week's big question: does my answer make sense? By the end you'll have a real checklist for it.

DAY 1

Positive, Negative & Absolute Value

What does the sign mean? Which number is greater? How far from zero?

NUMBERS TELL A STORY

A positive or negative sign isn't decoration — it tells you something real.

- +\$50 gained
- \$50 owed
- 8°F above zero
- 8°F below
- +100 ft elevation
- 100 ft below sea level

Before you calculate anything, ask what the sign actually means in the situation.

WHICH NUMBER IS GREATER?

Farther right on the number line means greater.



Every positive number is greater than every negative number. Zero is neither positive nor negative.

A TRAP WORTH KNOWING

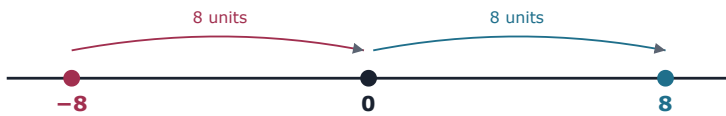
Is **-12** greater than **-5** ?

No — on the number line, -12 sits farther left.

Think about debt: would you rather owe \$5, or owe \$12?
A balance of $-\$5$ is better (greater) than a balance of $-\$12$.

ABSOLUTE VALUE

The distance a number is from zero.



$|-8| = 8$ and $|8| = 8$, because both are 8 units from zero. Absolute value is never negative — it doesn't mean "make the number positive." A temperature of -4°F doesn't become 4°F just because its absolute value is 4; it means -4°F is four degrees away from zero.

BEFORE WE MOVE ON

1. Compare two negative numbers.
2. Find an absolute value.
3. Interpret a negative value in context.

DAY 2

Operations & Order

Adding and subtracting with negatives · choosing the right operation · order of operations

ADDITION WITH NEGATIVES

Think of it as movement on the number line.

- **$-5 + (-3)$** : start at -5 , move 3 more negative $\rightarrow -8$
- **$-5 + 8$** : start at -5 , move 8 positive $\rightarrow 3$

SUBTRACTION WITH NEGATIVES

$$8 - (-3) = 11$$

Subtracting a negative removes a negative quantity, which increases the result. Think of it in context: if a \$3 loss is removed from your situation, your position improves by \$3.

LET'S WORK ONE TOGETHER

A bank account has \$45. A \$62 charge is processed. What's the new balance?

- **STOP:** \$45 balance, \$62 charge.
- **FIND:** the new balance.
- **CONNECT:** a charge decreases the balance.
- **TRY:** $45 - 62 = -17$.
- **CHECK:** a \$62 charge on a \$45 balance should go below zero — it does.

Answer: −\$17

MULTIPLYING & DIVIDING NEGATIVES

Same Signs → Positive

Different signs → negative. This applies to multiplication and division only.

It doesn't transfer to addition/subtraction — that's a different rule, and mixing them up is a common trap.

WHICH OPERATION DO I NEED?

Keywords are clues, not rules.

1. What quantities do I have?
2. What am I trying to find?
3. How are the quantities related?

"Total means add" and "each means multiply" are helpful clues — but the situation decides, not the word.

WHAT HAPPENS FIRST?

$6 + 3 \times 4$

- **Wrong:** $6+3=9$, then $9\times4=36$
- **Right:** $3\times4=12$, then $6+12=18$

A TRAP WORTH KNOWING

Does multiplication always come before division?

20 ÷ 5 × 2

No — multiplication and division have **equal priority**. Work left to right: $20 \div 5 = 4$, then $4 \times 2 = \mathbf{8}$ (not 2). Same for addition/subtraction: $12 - 5 + 3 \rightarrow 7 + 3 = \mathbf{10}$, not $12 - (5 + 3)$.

BEFORE WE MOVE ON

1. One integer operation
2. One order-of-operations problem
3. One error-analysis question

DAY 3

Estimation, Calculators & Reasonableness

About what should the answer be — before you trust the exact one?

ESTIMATE BEFORE YOU TRUST

$49 \times 21 \rightarrow \text{round to } 50 \times 20 \rightarrow \approx 1,000$

If a calculator shows 10,290, that should immediately look wrong. Estimating isn't random guessing — it's rounding to numbers you can work with easily, then reasoning from there.

WHICH ANSWER COULD BE RIGHT?

$$49 \times 21 = ?$$

A. 102.9 B. 1,029 C. 10,290 D. 102,900

$50 \times 20 \approx 1,000$, so only **B** is reasonable. You can eliminate three choices without calculating exactly — that's the whole point of estimating first.

A CALCULATOR ISN'T A CORRECTNESS CHECK

You meant 52×18 , but you typed 52×81 .

The calculator correctly answers the **wrong** calculation. It can't tell you that. Estimate first, then double-check what you actually entered.

THE WEEK 3 CHECKLIST

**Operation → Size → Sign →
Unit → Context**

Use this before accepting any answer as final — this week,
and every week after it.

BEFORE WE MOVE ON

**Why can't an incorrect calculator
result ever be "reasonable"?**

DAY 4

Apply, Analyze, Retry

Mixed practice, checking your own missed items, retrying after feedback.

TODAY'S PLAN

- Mixed retrieval: comparison, operations, order of operations, estimation, reasonableness
- When you miss a question, classify it — don't default to "I made a careless mistake" for everything
- Retry after reviewing an example or getting feedback
- Week 3 check, then a quick reflection: one strength, one developing skill, one error, what you did about it

TRANSITION TO WEEK 4

Fractions & Decimals

Same cycle: Learn → Practice → Check → Analyze → Retry → Apply. Keep using operation choice, order of operations, estimation, and reasonableness — they don't retire after this week.